

PROBLEM STATEMENT #30 | Smart Cities, Transport & Logistics Airport Lost Luggage Claim & Tracking Portal

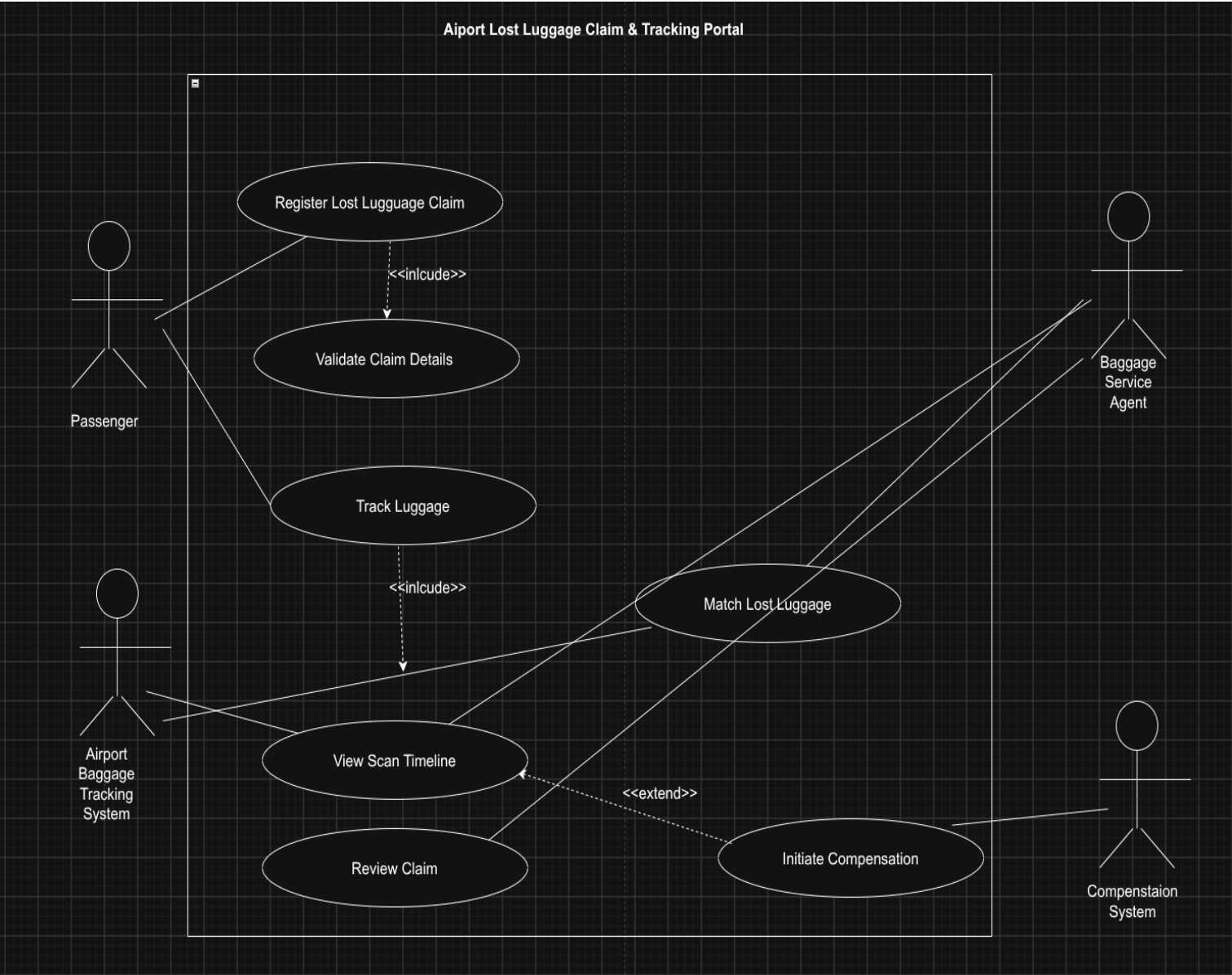
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1. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall cross-reference passenger lost baggage reports against unrouted luggage scan logs across all connected airport terminals.	High	Pass: A matching bag is identified and a recovery alert is generated. Fail: An incorrect bag-tag format is accepted.	Core functionality for locating lost luggage.
FR-002	Functional	The system shall allow a passenger to register a lost luggage claim using a bag tag number and flight details.	High	Pass: A valid claim is successfully created and assigned a claim ID. Fail: A claim is created with mandatory information missing.	Allows passengers to report missing luggage.
FR-003	Functional	The system shall display the luggage scan timeline and current tracking status to the passenger.	High	Pass: The system displays available scan locations and timestamps for a valid bag tag. Fail: Incorrect scan information is displayed.	Helps passengers monitor the luggage recovery process.
FR-004	Functional	The system shall allow a Baggage Service Agent to review and update the status of a lost luggage claim.	High	Pass: An authorized agent successfully updates the claim status and the updated status is stored. Fail: Unauthorized users can update the claim.	Enables staff to manage baggage recovery.
FR-005	Functional	The system shall initiate a passenger compensation workflow when a luggage claim meets the defined compensation conditions.	Medium	Pass: Eligible claims trigger a compensation request. Fail: An ineligible claim is automatically approved for compensation.	Supports automated handling of eligible compensation cases.
NFR-001	Non-Functional (Performance)	The baggage tracking timeline API must return complete multi-leg scan history in under 200 ms.	High	Pass: Benchmarking confirms the response target under simulated	Ensures fast tracking information and

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	Performance & Security)			peak load. Fail: The response exceeds 200 ms.	responsive service.
NFR-002	Non-Functional (Security)	The system shall restrict claim management functions to authenticated and authorized Baggage Service Agents.	High	Pass: Authorized agents can access claim management functions, while unauthorized users are denied access.	Protects passenger and baggage claim information.

2. UML Use-Case Diagram



3. Use-Case Flow Document

Use Case ID

UC-01

Use Case Name

Register Lost Luggage Claim

System

Airport Lost Luggage Claim & Tracking Portal

Primary Actor

Passenger:

Supporting System

Airport Baggage Tracking System

Preconditions

- 1.The Passenger has access to the Airport Lost Luggage Claim & Tracking Portal.
- 2.The Passenger has the required luggage and flight information.
- 3.The Passenger has a valid or identifiable bag tag number.
- 4.The Airport Baggage Tracking System is available for checking luggage information.

Main Success Scenario

- 1.The Passenger opens the **Airport Lost Luggage Claim & Tracking Portal**.
- 2.The Passenger selects **Register Lost Luggage Claim**.

3. The system displays the lost luggage claim form.
4. The Passenger enters the required claim details, including:
 - Passenger information
 - Bag tag number
 - Flight information
 - Contact information
5. The Passenger submits the lost luggage claim.
6. The system performs **Validate Claim Details** as an «include» use case.
7. The system verifies that all mandatory information has been provided and that the bag tag format is valid.
8. The system cross-references the submitted bag information with available luggage records from the Airport Baggage Tracking System.
9. The system successfully registers the lost luggage claim.
10. The system generates a unique claim ID.
11. The system stores the claim details for processing by the Baggage Service Agent.
12. The system displays a confirmation message and the claim ID to the Passenger.
13. The use case ends successfully.

Alternate Flow A1: Invalid or Missing Claim Details

This alternate flow begins at Step 7 of the Main Success Scenario.

1. The system detects that one or more required details are missing or invalid.
2. The system displays an error message identifying the incorrect or missing information.
3. The Passenger corrects or completes the required details.
4. The Passenger submits the claim again.
5. The system validates the updated claim details.

6.If the details are valid, the process continues from **Step 8** of the Main Success Scenario.

Alternate Flow A2: No Matching Luggage Record Found

This alternate flow begins at Step 8 of the Main Success Scenario.

- 1.The system cannot find a matching luggage record in the available baggage scan records.
- 2.The system informs the Passenger that the luggage has not yet been located.
- 3.The system still registers the lost luggage claim.
- 4.The claim is marked as **Pending Investigation**.
- 5.The Baggage Service Agent can later use **Match Lost Luggage** to investigate the claim.
- 6.The Passenger can subsequently use **Track Luggage** and **View Scan Timeline** to check for updated luggage information.
- 7.The use case ends successfully.

Postconditions

Success Postcondition

- A lost luggage claim is successfully registered.
- A unique claim ID is generated.
- The claim information is stored in the system.
- The claim is available for review by a Baggage Service Agent.

Alternate Postcondition

- If no luggage record is found, the claim remains registered with a **Pending Investigation** status.